

# Web Programming

# Web Programming

Rodrigo Martins Pagliares

# Introduction to HTML

# Advanced text formatting

Lecture: Advanced text formatting

Module: Introduction to HTML

# Introduction

# Introduction

- Here you'll learn about:
  - ✓ **Marking up quotations**
  - ✓ **Description lists**
  - ✓ **Computer code** and other related text
  - ✓ **Subscript** and **superscript**
  - ✓ **Contact** information
  - ✓ and more.

# Objective of the lecture

**Objective:** To learn how to use lesser-known **HTML** elements to mark up **advanced semantic features**.

# Description lists

# Description lists

- The **purpose of description lists** is to **mark up** a set of **items** and **their associated descriptions**.
- **Example:**
  - ✓ **Terms and definitions.**
  - ✓ **Questions and answers.**
  - ✓ ...

# The <dl>, <dt>, and <dd> elements

- <dl>
  - ✓ Description **list**.
- <dt>
  - ✓ Description **term**.
- <dd>
  - ✓ Description **definition**.

# The <dl>, <dt>, and <dd> elements

- |   |                                                                                                                              |
|---|------------------------------------------------------------------------------------------------------------------------------|
| 1 | soliloquy In drama, where a character speaks to themselves, representing their inner thoughts or feelings and in the process |
| 2 | relaying them to the audience (but not to other characters.)                                                                 |
| 3 | monologue In drama, where a character speaks their thoughts out loud to share them with the audience and any other           |
| 4 | characters present.                                                                                                          |
| 5 | aside In drama, where a character shares a comment only with the audience for humorous or dramatic effect. This is usually a |
| 6 | feeling, thought or piece of additional background information                                                               |

```
01 <dl>
02   <dt>soliloquy</dt>
03   <dd>In drama, where a character speaks to themselves, representing their inner thoughts or feelings
04       and in the process relaying them to the audience (but not to other characters.)</dd>
05   <dt>monologue</dt>
06   <dd>In drama, where a character speaks their thoughts out loud to share them with the audience
07       and any other characters present.</dd>
08   <dt>aside</dt>
09   <dd>In drama, where a character shares a comment only with the audience for humorous or
10       dramatic effect. This is usually a feeling, thought, or piece of additional background I
11       information.
12   </dd>
13 </dl>
```

# The <dl>, <dt>, and <dd> elements

**soliloquy**

In drama, where a character speaks to themselves, representing their inner thoughts or feelings and in the process relaying them to the audience (but not to other characters.)

**monologue**

In drama, where a character speaks their thoughts out loud to share them with the audience and any other characters present.

**aside**

In drama, where a character shares a comment only with the audience for humorous or dramatic effect. This is usually a feeling, thought or piece of additional background information.

Note that it is permitted to have a single term with multiple descriptions, for example:

**aside**

In drama, where a character shares a comment only with the audience for humorous or dramatic effect. This is usually a feeling, thought or piece of additional background information.

In writing, a section of content that is related to the current topic, but doesn't fit directly into the main flow of content so is presented nearby (often in a box off to the side.)

**Active learning:**  
Marking up a set of definitions

# Go to the link indicated below for the Active Learning statement

**Active learning: Marking up a set of definitions**

It's time to try your hand at description lists; add elements to the raw text in the *Input* field so that it appears as a description list in the *Output* field. You could try using your own terms and descriptions if you like.

If you make a mistake, you can always reset it using the *Reset* button. If you get really stuck, press the *Show solution* button to see the answer.

**Live output**

Bacon The glue that binds the world together. Eggs The glue that binds the cake together. Coffee The drink that gets the world running in the morning. A light brown color.

**Editable code**

Press Esc to move focus away from the code area (Tab inserts a tab character).

```
Bacon
The glue that binds the world together.
Eggs
The glue that binds the cake together.
Coffee
The drink that gets the world running in the morning.
A light brown color.
```

Reset Show solution

<https://mzl.la/39cHxZn>

# Quotations

# The `<blockquote>` element and the `cite` attribute

- **Blockquotes**

- ✓ If a section of block level content is quoted from somewhere else, you should:

- Wrap it inside a `<blockquote>` element to signify this.
    - Include a URL pointing to the source of the quote inside a `cite` attribute.

- The section of block level content may a paragraph, multiple paragraphs, a list, etc..

# The `<blockquote>` element and the `cite` attribute

```
1 <p>The <strong>HTML <code>&lt;blockquote&gt;</code> Element</strong>  
2 (or <em>HTML Block Quotation Element</em>) indicates that the enclosed text is an extended quotation.</p>
```



Wrap it !

```
1 <blockquote cite="https://developer.mozilla.org/en-US/docs/Web/HTML/Element/blockquote">  
2   <p>The <strong>HTML <code>&lt;blockquote&gt;</code> Element</strong> (or <em>HTML Block Quotation Element</em>)  
3   indicates that the enclosed text is an extended quotation.</p>  
4 </blockquote>
```



Output

The **HTML `<blockquote>` Element** (or *HTML Block Quotation Element*) indicates that the enclosed text is an extended quotation.

# The <q> element and the <cite> attribute

Used for inline quotations

```
1 <p>The quote element — <code>&lt;q&gt;</code> — is  
2   <q cite="https://developer.mozilla.org/en-US/docs/Web/HTML/Element/q">  
3     intended for short quotations that don't require paragraph breaks.</q>  
4 </p>
```

The quote element — <q> — is "intended for short quotations that don't require paragraph breaks".

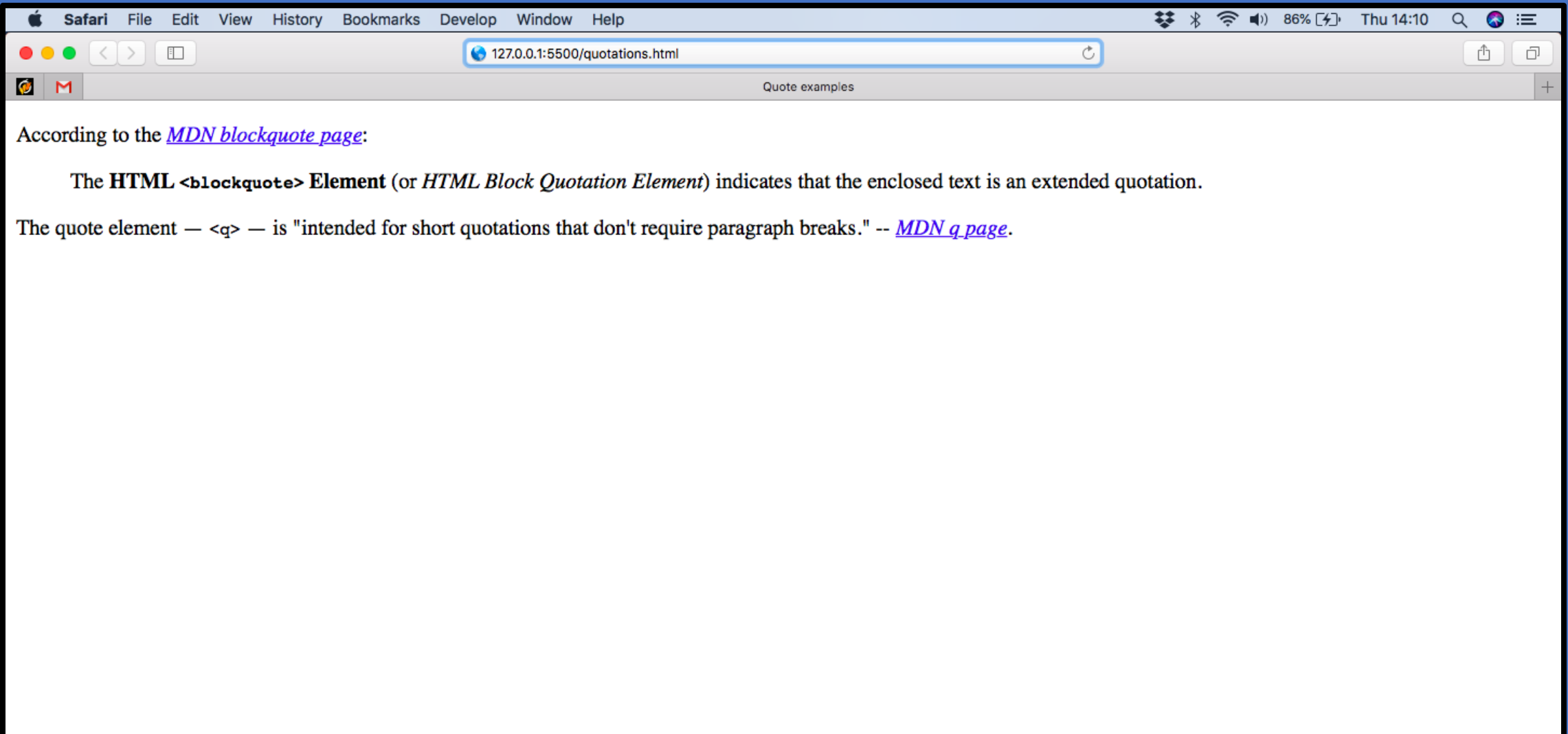
# The `<cite>` element and the `cite` attribute

- The **content** of the `cite` **attribute** sounds useful, but unfortunately **browsers, *screenreaders*, etc. don't really do much** with it.
- There is **no way** to get the **browser** to **display** the **contents** of `cite`, **without writing your own solution** using **JavaScript** or **CSS**.
- There is a `<cite>` **element**, but this is **meant to contain the title** of the **resource being quoted**
  - ✓ E.g. the **name** of the **book**.

# The <cite> element and the cite attribute

- There is no reason however why you couldn't link the text inside <cite> to the quote source in some way.

```
1 <p>According to the <a href="https://developer.mozilla.org/en-US/docs/Web/HTML/Element/blockquote">
2   <cite>MDN blockquote page</cite></a>:
3 </p>
4
5 <blockquote cite="https://developer.mozilla.org/en-US/docs/Web/HTML/Element/blockquote">
6   <p>The <strong>HTML <code>&lt;blockquote&gt;</code> Element</strong> (or <em>HTML Block Quotation
7     Element</em>) indicates that the enclosed text is an extended quotation.</p>
8 </blockquote>
9
10 <p>The quote element — <code>&lt;q&gt;</code> — is
11   <q cite="https://developer.mozilla.org/en-US/docs/Web/HTML/Element/q">intended for short quotations that
12   don't require paragraph breaks </q>
13   --<a href="https://developer.mozilla.org/en-US/docs/Web/HTML/Element/q"> <cite>MDN q page</cite></a>.
14 </p>
```



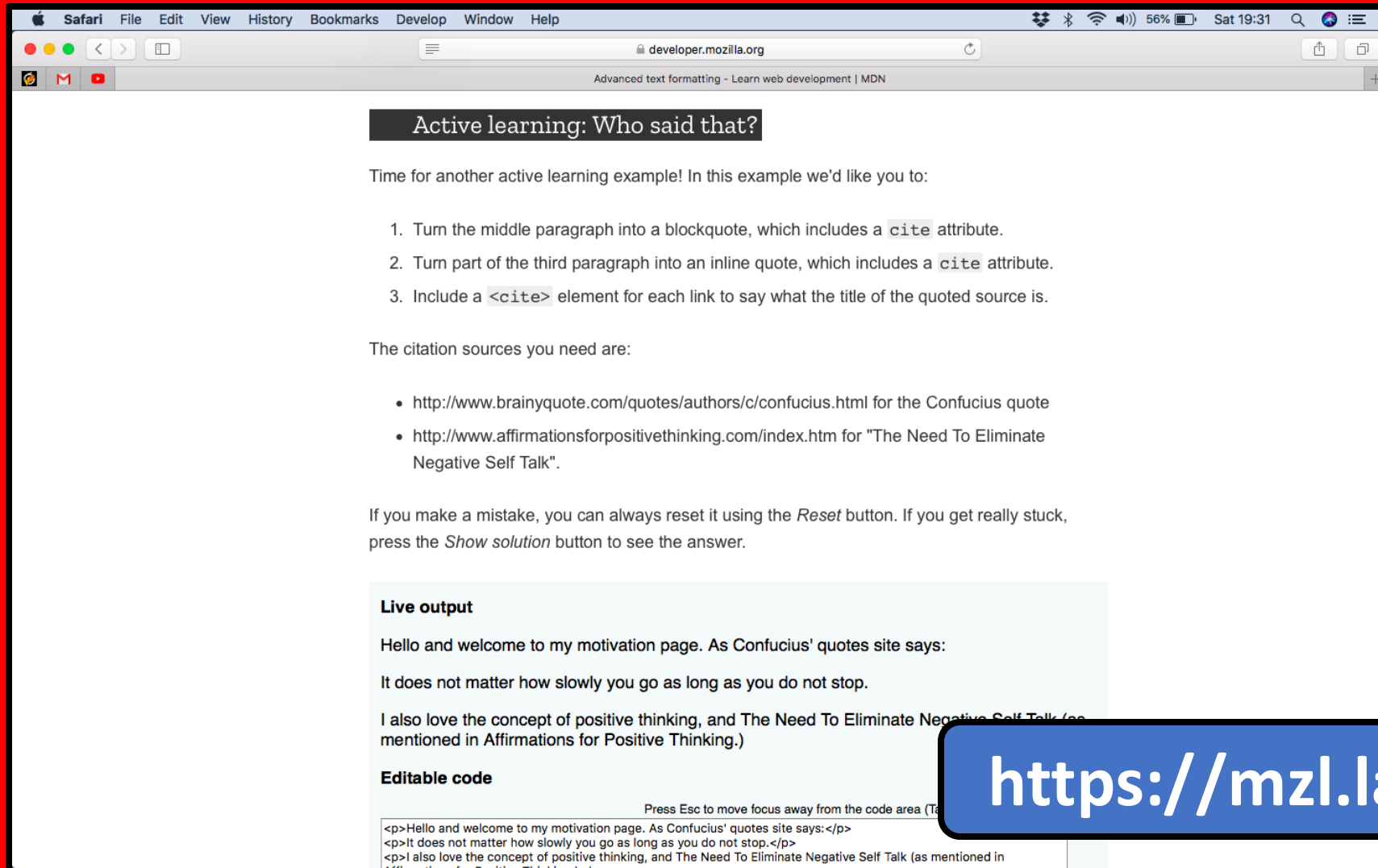
Running the example locally (127.0.0.1) at  
LiverServer

# Rationale

- The **attributes** `cite`, although **not visible**, are **useful** due to the **semantics** they **provide** to the `blockquote` and `q` elements.
- The `<cite>` element usually represents the `title` of the resource being cited.
- **Good practice:** wrap the **quotation** in anchor **element** (`<a>`).

**Active learning:**  
Who said that?

# Go to the link indicated below for the Active Learning statement



The screenshot shows a Safari browser window with the address bar at `developer.mozilla.org`. The page title is "Advanced text formatting - Learn web development | MDN". The main heading is "Active learning: Who said that?". The text says: "Time for another active learning example! In this example we'd like you to:" followed by a list of three tasks:

1. Turn the middle paragraph into a blockquote, which includes a `cite` attribute.
2. Turn part of the third paragraph into an inline quote, which includes a `cite` attribute.
3. Include a `<cite>` element for each link to say what the title of the quoted source is.

The citation sources you need are:

- <http://www.brainyquote.com/quotes/authors/c/confucius.html> for the Confucius quote
- <http://www.affirmationsforpositivethinking.com/index.htm> for "The Need To Eliminate Negative Self Talk".

If you make a mistake, you can always reset it using the *Reset* button. If you get really stuck, press the *Show solution* button to see the answer.

**Live output**

Hello and welcome to my motivation page. As Confucius' quotes site says:

It does not matter how slowly you go as long as you do not stop.

I also love the concept of positive thinking, and The Need To Eliminate Negative Self Talk (as mentioned in Affirmations for Positive Thinking.)

**Editable code**

Press Esc to move focus away from the code area (Tab)

```
<p>Hello and welcome to my motivation page. As Confucius' quotes site says:</p>
<p>It does not matter how slowly you go as long as you do not stop.</p>
<p>I also love the concept of positive thinking, and The Need To Eliminate Negative Self Talk (as mentioned in
```

<https://mzl.la/3ajxcLk>

# Abbreviations

# The <abbr> element and its title attribute

- Used to **wrap around** an **abbreviation** or **acronym** and **provide** a **full expansion** of the **term** (included inside a **title attribute**).

```
1 <p>We use <abbr title="Hypertext Markup Language">HTML</abbr> to structure our web
2   documents.</p>
3 <p>I think <abbr title="Reverend">Rev.</abbr> Green did it in the kitchen with the
4   chainsaw.</p>
```

- The **expansion** will **appear** in a **tooltip** when the **term** is **hovered over**.

We use HTML to structure our web documents.

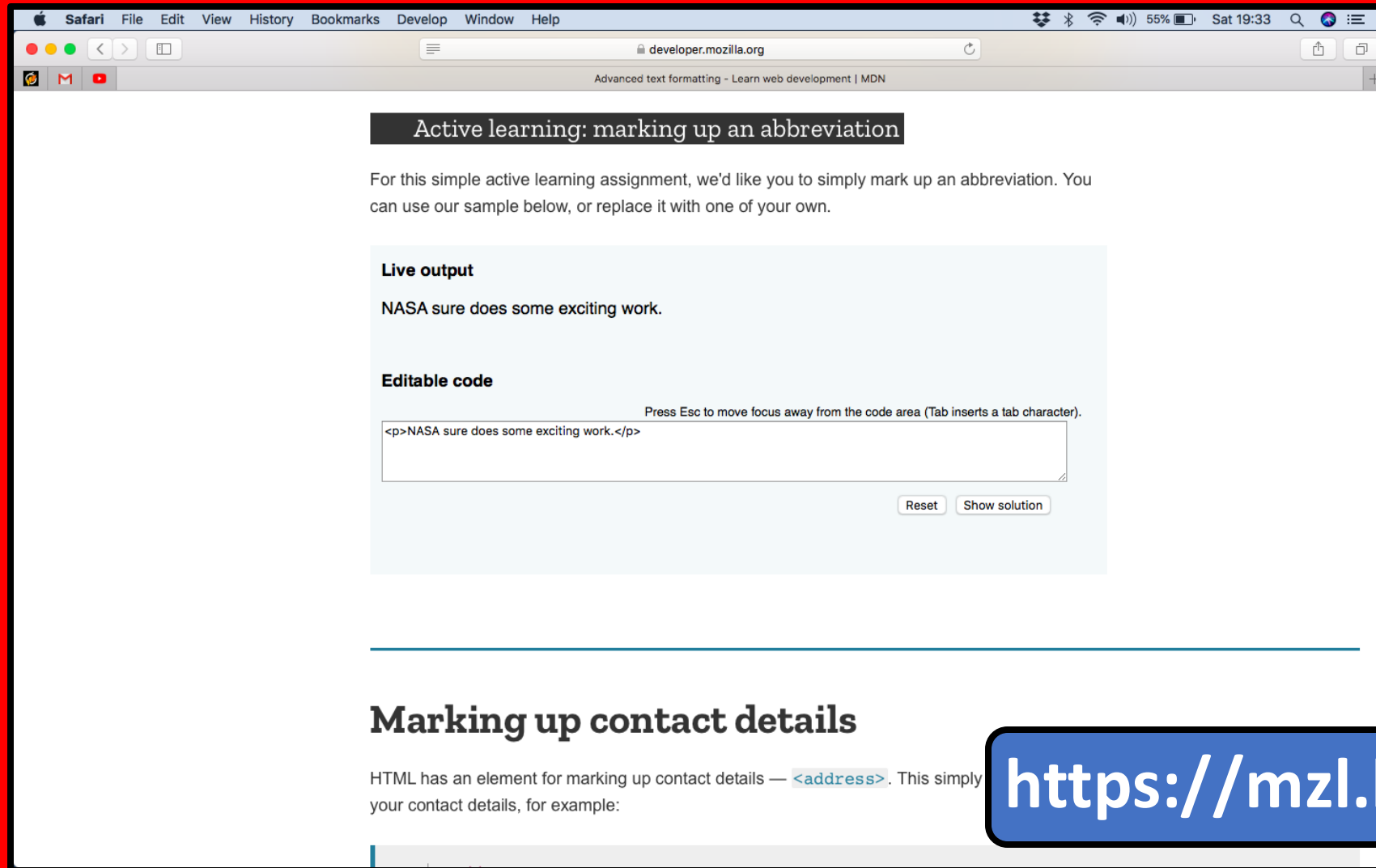
I think Rev. Green did it in the kitchen with the chainsaw.

# Note

**Note:** There is another element, `<acronym>`, which basically does the same thing as `<abbr>`, and was intended specifically for acronyms rather than abbreviations. This however has fallen into **disuse** — it wasn't supported in browsers as well as `<abbr>`, and has such a similar function that it was considered pointless to have both. **Just use `<abbr>` instead.**

**Active learning:**  
Marking up an abbreviation

# Go to the link indicated below for the Active Learning statement



The screenshot shows a Safari browser window with the address bar at `developer.mozilla.org`. The page title is "Advanced text formatting - Learn web development | MDN". The main content area has a dark header "Active learning: marking up an abbreviation". Below it, a paragraph explains the task: "For this simple active learning assignment, we'd like you to simply mark up an abbreviation. You can use our sample below, or replace it with one of your own." The interface includes a "Live output" section showing "NASA sure does some exciting work." and an "Editable code" section with a text area containing `<p>NASA sure does some exciting work.</p>`. A hint above the code area says "Press Esc to move focus away from the code area (Tab inserts a tab character)". At the bottom of the code area are "Reset" and "Show solution" buttons. Below the exercise, a section titled "Marking up contact details" begins with the text "HTML has an element for marking up contact details — `<address>`. This simply your contact details, for example:".

<https://mzl.la/2vsmCTu>

**Marking up contact details**

# The <address> element

```
1 <address>
2   <p>Chris Mills, Manchester, The Grim North, UK</p>
3 </address>
```

```
1 <address>
2   <p>Page written by <a href=" ../authors/chris-mills/">Chris Mills</a>.</p>
3 </address>
```

```
01 <address>
02   <p>
03     Chris Mills<br>
04     Manchester<br>
05     The Grim North<br>
06     UK
07   </p>
08
09   <ul>
10     <li>Tel: 01234 567 890</li>
11     <li>Email: me@grim-north.co.uk</li>
12   </ul>
13 </address>
```

ok, if the linked page contained the contact information

**Superscript & subscript**

# The <sup> and <sub> HTML elements

```
1 <p>My birthday is on the 25<sup>th</sup> of May 2001.</p>  
2 <p>Caffeine's chemical formula is C<sub>8</sub>H<sub>10</sub>N<sub>4</sub>O<sub>2</sub>.</p>  
3 <p>If x<sup>2</sup> is 9, x must equal 3 or -3.</p>
```

The **output** of this code looks like so

My birthday is on the 25<sup>th</sup> of May 2001.

Caffeine's chemical formula is C<sub>8</sub>H<sub>10</sub>N<sub>4</sub>O<sub>2</sub>.

If x<sup>2</sup> is 9, x must equal 3 or -3.

# Representing computer code

# The <code>, <pre> and <var> elements

- **Elements available for marking up computer code using HTML:**

- ✓ <code>

- For marking up **generic pieces of computer code**.

- ✓ <pre>

- For **retaining whitespace** (generally **code blocks**).

- ✓ <var>

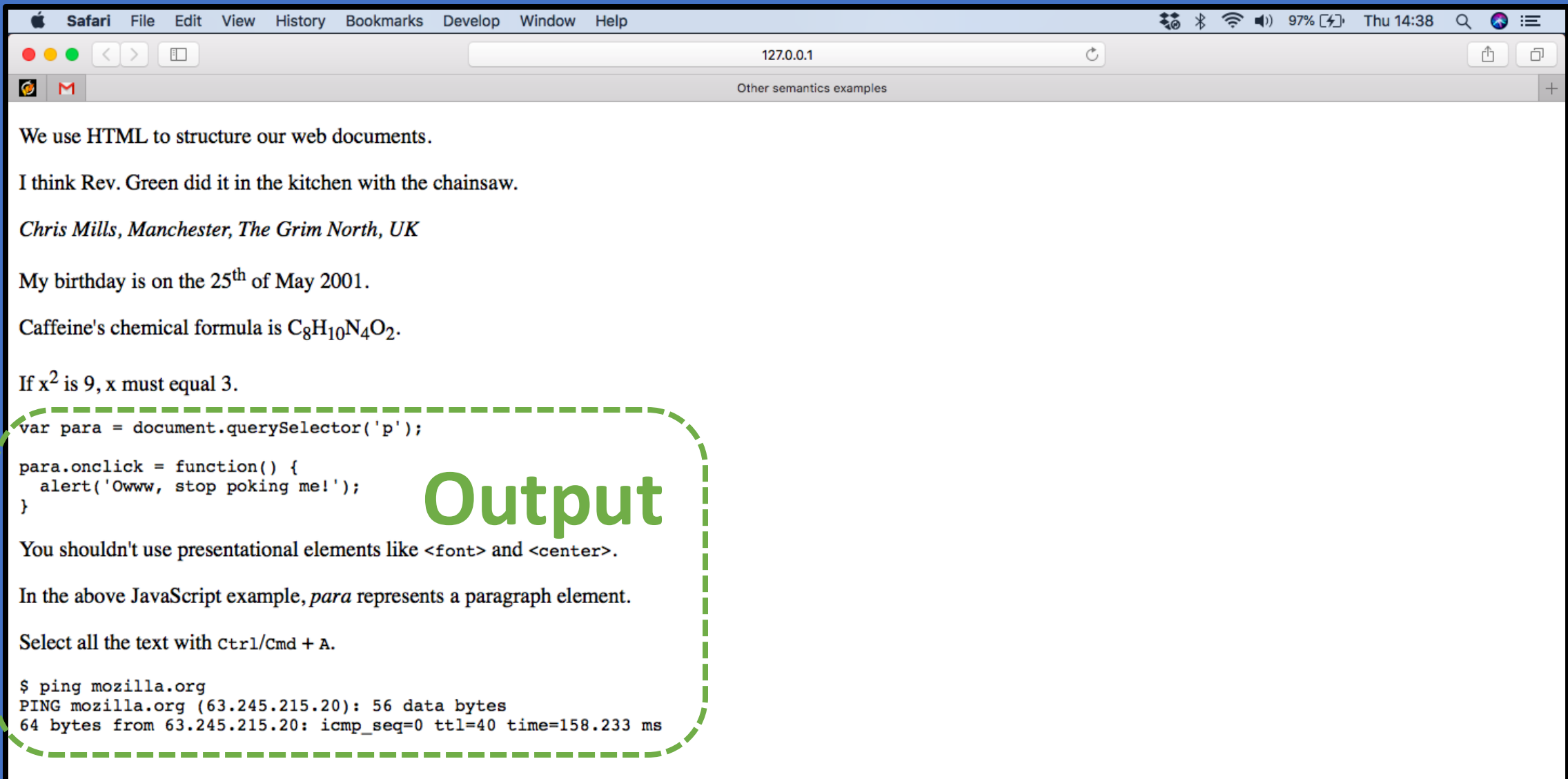
- For specifically **marking up variable names**.

# The `<kbd>` and `<samp>` elements

- More elements available for marking up computer code using HTML.
  - ✓ `<kbd>`
    - For marking up keyboard (and other types of) input entered into the computer.
  - ✓ `<samp>`
    - For marking up the output of a computer program.

# Example

```
01 <pre>
02   <code>var para = document.querySelector('p'); para.onclick = function() { alert('Owww, stop poking me!'); }</code>
03 </pre>
04
05 <p>You shouldn't use presentational elements like <code>&lt;font&gt;</code> and <code>&lt;center&gt;</code>.</p>
06
07 <p>In the above JavaScript example, <var>para</var> represents a paragraph element.</p>
08
09 <p>Select all the text with <kbd>Ctrl</kbd>/<kbd>Cmd</kbd> + <kbd>A</kbd>.</p>
10
11 <pre>$
12   <kbd>ping mozilla.org</kbd>
13   <samp>PING mozilla.org (63.245.215.20): 56 data bytes 64 bytes from 63.245.215.20: icmp_seq=0 ttl=40
14     time=158.233 ms</samp>
15 </pre>
```



Running the example locally (127.0.0.1) at  
LiverServer

**Marking up times and dates**

# The <time> element

- Marks up times and dates in a machine-readable format.

```
<time datetime="2016-01-20">20 January 2016</time>
```

# Why is <time> useful?

- There are **many different ways** that **humans write down dates**:
  - ✓ 20 January 2016.
  - ✓ 20th January 2016.
  - ✓ Jan 20 2016.
  - ✓ 20/01/16.
  - ✓ 01/20/16.
  - ✓ The 20th of next month.
  - ✓ 20e Janvier 2016.
  - ✓ 2016年1月20日.
  - ✓ ...

# Why is `<time>` useful?

- These **different forms** (previous slide) **cannot** be easily **recognized** by **computers**.
- The `<time>` **element** allows you to **attach** an **unambiguous, machine-readable time/date** for this **purpose**.

# Options

**<!-- Standard simple date -->**

`<time datetime="2016-01-20">20 January 2016</time>`

**<!-- Just year and month -->**

`<time datetime="2016-01">January 2016</time>`

**<!-- Just month and day -->**

`<time datetime="01-20">20 January</time>`

**<!-- Just time, hours and minutes -->**

`<time datetime="19:30">19:30</time>`

**<!-- You can do seconds and milliseconds too! -->**

`<time datetime="19:30:01.856">19:30:01.856</time>`

**<!-- Date and time -->**

`<time datetime="2016-01-20T19:30">7.30pm, 20 January 2016</time>`

**<!-- Date and time with timezone offset-->**

`<time datetime="2016-01-20T19:30+01:00">7.30pm, 20 January 2016 is 8.30pm in France</time>`

**<!-- Calling out a specific week number-->**

`<time datetime="2016-W04">The fourth week of 2016</time>`

**Test your skills**

Safari File Edit View History Bookmarks Develop Window Help

developer.mozilla.org

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# Test your skills: Advanced HTML text

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## On this Page

- [Advanced HTML text 1](#)
- [Advanced HTML text 2](#)
- [Assessment or further help](#)

This aim of this skill test is to assess whether you've understood our [Advanced text formatting](#) article.

**Note:** You can try out solutions in the interactive editors below, however it may be helpful to download the code and use an online tool such as [CodePen](#), [jsFiddle](#), or [Glitch](#) to work on the tasks.

If you get stuck, then ask us for help — see the [Assessment or further help](#) section at the bottom of this page.

## Advanced HTML text 1

In this task we want you to turn the provided animals and their definit

## Related Topics

[Complete beginners start here!](#)

- ▶ [Getting started with the Web](#)

[HTML — Structuring the Web](#)

- ▾ [Introduction to HTML](#)
  - [Introduction to HTML overview](#)
  - [Getting started with HTML](#)
  - [What's in the head? Metadata in HTML](#)

<https://mzl.la/2wfwK2e>

**Questions?**